MIL-F-60349B (MU)
1 February 1970
SUPERSEDING
MIL-F-60349A (MU)
16 August 1968

MILITARY SPECIFICATION

FUZE, POINT DETONATING, M48A3, LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for the M48A3 Point Detonating Artillery Ammunition Fuze.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect of date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-I-45607	- Inspection Equipment, Supply and Maintenance Of
MIL-D-46256	- Detonator, Stab, M24, Loading, Assembling and Packing
MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes; General Specification For

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attrubutes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

8798219 - Fuze, Point Detonating, M48A3, Assembly
8798234 - Screw, Detonator Retaining

PUBLICATIONS

ARMY

ET-8798219 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-The materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and Assemblies.-The parts and assemblies shall comply with all requirements of Drawing (dwg.) 8798219 and with all requirements specified in the applicable specifications.

3.3 Jolt and Jumble.-The fuze assembly shall comply with the requirements of MIL-STD-331, tests 101 and 102.

3.4 Transportation Vibration.-The fuze assembly shall comply with the requirements of MIL-STD-331, test 104.

3.5 Arming and Propagation Test.-The loaded fuze assembly, dwg. No. 8798219, shall arm and function high order when subjected to this test. The test shall be performed in accordance with 4.2.3.3.

3.6 Proving Ground

3.6.1 Premature Bursts.-There shall be no premature bursts in the gun bore or in flight before impact when tested as specified in 4.2.3.4.

3.6.2 Fuctioning

3.6.2.1 Superquick.-The fuzes shall cause the projectile to function high order and shall show no evidence of reverse functioning (see 6.4) when tested as specified in 4.2.3.4.

3.6.2.2 Delay.-The Fuze shall cause the projectile to function high order, and the Fuze shall show no evidence of reverse functioning.

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3.7 Workmanship.-All parts and assemblies shall be loaded and assembled in a thorough, workmanlike manner and all manufacturing, processing, and assembling operations shall be correctly performed. All parts shall be free of burrs, sharp edges, cracks, dirt, grease, dust, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible, and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and the certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Parts and parts assemblies of one lot interfix number from one manufacturer.

b. Delay plunger assemblies of one lot interfix number from one manufacturer.

c. Detonator assemblies of one lot interfix number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Screw, Detonator, Retaining (see dwg. 8798234).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Pitch diameter of thread, min. ..	Gage	01001
Minor:	AQL 1.5 Percent		
201.	Total length	Gage	01002
202.	Pitch diameter of thread, max. ..	Gage	01003
203.	Evidence of poor workmanship (see 3.7)	Visual	01004

4.2.2.2 Head Assembly Loading (see dwg. 8798232 covering a detail of dwg. 8798219).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Detonator missing or inverted ...	Visual	02001
102.	Detonator cushion missing	Visual	02002
103.	Detonator retaining screw not fully seated	Visual	02003
104.	Detonator retaining screw not staked	Visual	02004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	02005

4.2.2.3 Body Assembly Loading (see dwg. 9234457).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Tape missing or does not completely cover flash hole	Visual	06001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	06002

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4.2.2.4 Assembly, prior to assembling head assembly and bottom closing screw (see dwg. 8798219).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Delay plunger centrifugal plunger pin in armed condition	Visual	03001
2.	Interrupter does not completely obstruct the flash hole	Visual	03002
Major:	AQL 0.65 Percent		
101.	Foreign matter in flash hole or in delay plunger assembly	Visual	03003
102.	Flash tube stake missing	Visual	03004
103.	Foreign matter in flash tube cavity	Visual	03005
104.	Delay plunger assembly missing ..	Visual	03006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	03007

4.2.2.5 Assembly (see dwg. 8798219).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.0 Percent		
101.	Total length, above max.	Gage	04002
102.	Bottom closing screw not below flush, min.	Gage	04003
103.	Ogive covers any part of setting sleeve	Visual	04004
104.	Body thread damaged	Visual	04005
105.	Delay time not specified on fuze.	Visual	04006
106.	Fuze not set superquick	Visual	04007
107.	Crack or puncture at ogive head stake or dimples	Visual	03008
108.	Ogive head stake missing	Visual	04009
109.	Delay plunger assembly inverted .	Visual	04001
110.	Tape missing or does not completely cover bottom closing screw (applicable as indicated on Note 12 of dwg. 8798219)	Visual	04014

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Minor:	AQL 1.5 Percent		
201.	Firing pin closing disc missing or damaged	Visual	04010
202.	Protective coatings missing from markings	Visual	04011
203.	Marking missing, misleading, or unidentifiable	Visual	04012
204.	Evidence of poor workmanship (see 3.7)	Visual	04013

4.2.3 Testing

4.2.3.1 Jolt and Jumble (see 3.3).-Critical defects Code No. 05020 and 05021.

4.2.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of thirty (30) loaded fuze assemblies shall be selected from each lot and subjected to the jolt test specified in MIL-STD-331, test 101. After this test, the assemblies shall not be disassembled but shall be subjected to the jumble test, MIL-STD-331, test 102. The assemblies shall then be disassembled to determine compliance with MIL-STD-331. As a minimum inspection requirement for Critical defects, no M1-delay plunger or body interrupter shall be in an armed or partially armed condition. If any assembly fails to comply with the requirements of MIL-STD-331, the lot shall be rejected. Test equipment shall be in accordance with 4.2.4. Assemblies subjected to these tests shall not be returned to the lot.

4.2.3.1.2 After three consecutive lots have met the criteria specified in 4.2.3.1.1, a sample of ten (10) loaded fuze assemblies shall be selected from each lot for jolt and jumble tests. Except for sample size, the test procedure and acceptance criteria shall be those specified in 4.2.3.1.1.

4.2.3.2 Transportation Vibration (see 3.4).- Critical defect Code No. 05022 for safety deficiencies and Major defect, Code No. 05023, for operational deficiencies.

4.2.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of thirty (30) loaded fuze assemblies shall be selected from each lot and subjected to the transportation vibration test specified in MIL-STD-331, test 104. The assemblies shall then be disassembled to determine compliance with MIL-STD-331. As a minimum inspection requirement for Critical defects, no M1-delay plunger, or body interrupter shall be in an armed or partially armed condition.

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As a minimum inspection requirement for Major defects, no metal parts components shall be broken. If any assembly fails to comply with the requirements of MIL-STD-331, the lot shall be rejected. Test equipment shall be in accordance with 4.2.4. Assemblies subjected to this test shall not be returned to the lot.

4.2.3.2.2 After three consecutive lots have met the criteria specified in 4.2.3.2.1, a sample of ten (10) loaded fuze assemblies shall be selected from each lot for transportation vibration tests. Except for sample size, the test procedure and acceptance criteria shall be those specified in 4.2.3.2.1.

4.2.3.3 Arming and Propagation Test - Major defect - Arming, Code No. 05024 - Major defect - Propagation, Code No. 05025.-The test shall be performed as specified in4.3.2 using equipment in accordance with 4.2.4. Fifty (50) loaded fuze assemblies shall be randomly selected from each lot. All assemblies shall be tested in the delay position. The test shall be conducted for information only.

4.2.3.4 Proving Ground

4.2.3.4.1 Premature burst (see 3.6.1) - Critical defect.- Any premature burst shall be cause for rejection of the lot.

4.2.3.4.2 Functioning (see 3.6.2) - Major defect.-Thirty five (35) loaded fuze assemblies shall be selected from each lot and tested as specified in 4.3.3. The acceptance criteria and the number of fuzes to be tested shall be as specified in Table I.

TABLE I

Weapon	Target	Target Distance	Sample Size S.Q.	Sample Size Delay
4.2 Inch Mortar	Ground Impact	Convenient	25	
4.2 Inch Mortar	Ground Impact	Convenient		10

ACCEPTANCE CRITERIA

Defect Superquick	Accept	Reject	Code No.
Premature Burst	0	1	10001
Duds	1	2	10002
Reverse Functions	3	4	10003

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Delay	Accept	Reject	Code No.
Premature Burst	0	1	10004
Duds	Info	only	10005
Reverse Functions	Info	only	10006

4.2.3.5 Check-test for deterioration of detonator, Major defect, Code No. 11001.-If the total elapsed time between the acceptance of any detonator lot and assembly of that lot into fuze lots exceeds two (2) years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be check-tested as specified in MIL-D-46256 immediately prior to detonator assembly into fuzes. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the fuze. The test shall be conducted at Government expense without cost to the supplier who loaded the detonator or the supplier assembling the detonator into the fuze, and shall constitute a basis for rejection against either supplier except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. while the detonator lot is under the jurisdiction of either contractor.

4.2.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-8798219 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment. The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

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4.3 Test Methods and Procedures

4.3.1 Jolt and Jumble.-The loaded fuze assemblies shall be subjected to the jolt test in accordance with MIL-STD-331, Test 101. After completion of the jolt test, the fuzes shall be subjected to the jumble test in accordance with MIL-STD-331, Test 102. The jumble test shall be conducted utilizing a jumble protective fixture as specified in 4.2.4.

4.3.2 Arming and Propagation Test.-The loaded fuze assemblies shall be armed by spinning at no more than 3000 rpm in a suitable test fixture (see 4.2.4). The assemblies shall then be impact detonated by dropping on a hard surface. The assemblies shall be dropped while spinning. Assemblies found defective shall be examined and the cause of failure shall be annotated on the inspection records for the lot.

4.3.3 Functioning.-The test fuzes shall be assembled into 4.2 Inch Mortar Smoke Projectiles using appropriate thread adapters and tested according to Table II.

TABLE II

	4.2 Inch Mortar
Recommended Model	M30
Recommended Cartridge	M328A1
Recommended Charge	7 Increments
Muzzle Velocity (fps)	400
Pressure (psi)	- - - - - -
Spin (rpm)	3400

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.-Packaging, packing and marking shall be in accordance with the requirements and inspection procedures specified in MIL-P-60412.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all

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information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Reverse functioning.-Reverse functioning is defined as superquick (SQ) functioning of the fuze when set in the "Delay" position or delay functioning of the "SQ" position.

6.5 Cost of detonator deterioration check-test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the detonation check test.

6.6 Identification of international standardization agreements.-Certain provisions of Section 3, REQUIREMENTS OF this specification are the subject of international standardization agreement ABC-ARMY-STD-56. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective department standardization offices so that appropriate action may be taken respecting the international agreement concerned.

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6.7 Submission of test data and data cards.-In addition to the normal distribution of records, when the fuze is procured by the Department of the Army, a copy of all test data and data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND7 and DJ5
Dover, New Jersey 07801

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-787

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